ALA:2

Python Deep Dive: Interview Intensive

1. What are the rules for naming variables in Python?
Answer:

- Must start with a letter or underscore (_)
- Cannot start with a number
- Can contain letters, numbers, and underscores
- Case-sensitive (age ≠ Age)
- Cannot use Python keywords (if, for, class)

2. Which of the following are valid variable names?
a = 10
_name = 20
2value = 30
user-name = "John"
total_marks = 50
Answer:
Valid → a, _name, total_marks
Invalid → 2value, user-name

3. Can variable names contain spaces? Why or why not?
Answer:
No, spaces are not allowed because Python treats them as separators between words.

4. What happens if you use a keyword as a variable name?
Answer:
It will give a syntax error
Example:
if = 10 # Error

5. Why are variable names case-sensitive in Python?
Answer:
Because Python treats uppercase and lowercase letters differently.
Example:
age = 20
Age = 25
print(age) # 20
print(Age) # 25

6. What is snake_case in Python?
Answer:
A naming style where words are separated by underscores
Example:
user_age = 21
total_price = 1000

7. Can variable names include Unicode characters?
Answer:
Yes, Python allows Unicode names, but it is not recommended for readability.
Example:
नाम = "Soham"

8. What is the difference between camelCase and snake_case?
Answer:

- camelCase → userName
- snake_case → user_name (recommended in Python)

9.What happens if two variables have the same name?

- **Answer:**

If two variables have the same name in Python, the latest assigned value overwrites the previous value. This means the variable will only store the most recent value, and the old value will be lost.
Example:
x = 10
x = 20
print(x)
Output:
20

10. Which is better variable naming and why?
x = 100
total_salary = 100
Answer:
total_salary is better because it is clear and meaningful